Bioinformatics_Bootcamp_Assessment_01 – Raima Khan

Answer#01:

For determining whether the DNA sequence has been recorded before, NCBI Genbank, ClinVar etc, I would suggest to use as the primary databases to check that DNA sequence out.

NCBI GenBank is the most comprehensive database of nucleotide seq. which is easy to access publicly. Provides annotated seq. data including organism of origin, gene name, location of gene, protein product, or associated publications. Other than these researchers can also use BLAST, basic local alignment search tool to compare query seq. against whole collection of genbank nucleotide. If the seq. exists, the BLAST would return an identical or highly similar matches with detailed information about the source organism, gene function and the literature relevant to that DNA seq. , as the ev value and parent percentage also helpful in this process which would be get from the genbank nucleotide.

Answer#02:

To compare the disease causing bacterium genome with related bacterial species to identify conserved genes, I would suggest the following:

NCBI Genome DataBases provides complete and draft genome assemblies for thousands of bacterial species. It offers easy access to fully annotated genomes, enabling the scientists to get/ download complete genomic seq. of related bacterial strains. The databases include the detailed information about genome size, GC content, gene count and also the functional annotation. **NCBI GenBank** remains the very essential for accessing individual gene seq. and its annotations. By retrieving seq. from related bacterial species the researchers can perform pairwise or multiple seq. alignment to identify the conserved regions. And also **KEGG**, Kyoto Encyclopedia of genes and genome is the great for understanding the functional context of conserved genes. It provides pathways maps which show how genes can interact in metabolic and signaling the pathways. The conserved genes are involved in essential pathways can be identified through the KEGG pathway analysis. These databases enable comprehensive comparative genomic assisting to identify genes necessary for the bacterial survival and virulence that could be served as potential drug targets.

Answer #03:

To investigate whether similar proteins exists in other species after isolating the protein from a newly identified microorganism, **UNIPROT,** Universal protein resource is the most comprehensive protein database that combines information from swiss prot, TrEMBL and PIR. To provide the seq. data, functional annotations, domain information and cross references to the other databases. The researchers can search uniport using BLASTp, protein BLAST to find similar proteins across all species. It also offers information about the protein function, subcellular localization, post translational modification and associated diseases or phenotypes. **NCBI Protein DataBase** serves as complementary resource which contain the protein seq. derived from GenBank coding the seq. It includes proteins from all domains of life and viruses that make it valuable to identify homologs in diverse organisms. The databases provides the accession numbers, seq. length, organism information and connects with the relevant literature. **RCSB Protein Data Bank,** is also the open access archive which is used to visualized the 3D structure of protein that assist to understand the function and evolutionary relationships through structural homology that even when seq similarity is weak.

Answer#04:

To determine the function of the unknown protein seq, here is the recommended database approaches; **UniProt** should be the first resource consulted. Using BLASTp against UniProtKB, the researchers could find the homologous proteins with the known functions. It provides comprehensive annotations including the molecular functions, biological processes, cellular components and the relevant literature references. The Swiss-Prot section of it contains manually curated and reviewed entries that offer the highest confidence annotations. **Gene Ontology (GO),** resources provide the standardized terminology for describing the protein functions. The annotations help systematically to classify the molecular functions of protein, biological processes and cellular component which facilitates in comparisons across organisms. These comprehensive approaches maximize the chances of accurately predicting function from seq alone.

Answer #05:

Firstly, I would examine the composition of seq to determine its type. The DNA contains only A, T, C, and G, while RNA contains U instead of T. The Protein seq consists of 20 different amino acids which make it easily distinguishable. Then I would also check seq length and look for any headers or annotations that might provide clues about the origin.

Secondly, to identify the open reading frames and translate the seq in all the 6 reading frames by using tool like NCBI's ORFfinder,. The presence of long ORFs without the premature stop codons suggests protein-coding DNA, while the absence might indicate the non-coding RNA.

Thirdly, I would perform BLASTn against the GenBank to find matching DNA seq. If unsuccessful, then would use the BLASTx from NCBI nucleotide database to translate the nucleotide seq and compare it against the protein databases. For protein seq, there would be direct use BLASTp against UniProt or NCBI protein database to find the homologous proteins with known functions.

Fourthly, to identify the conserved domains and protein families, I would use InterPro. UniProt would provide comprehensive functional annotations while KEGG could place the protein in relevant metabolic or signaling pathways.

Furthermore, I would evaluate BLAST results by examining E-values and bit scores, ensuring only statistically significant matches inform my conclusions.

Lastly, I would compile all evidence from multiple databases to confidently confirm the seq type, predict function and identify the source organism.